Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/03/2005
PI Name:	Wray, Gregory Ph.D.		
Project Title:	Evolution Of Endo16 Promoter Function In Camarodont Sea Urchins		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
PI Email:	gwray@duke.edu	**Fax:**	FY 919-660-2793
PI Organization Type:	UNIVERSITY	**Phone:**	919-684-6696
Organization Name:	Duke University		
PI Address 1:	Department of Biology		
PI Address 2:	Box 90338		
PI Web Page:	http://www.biology.duke.edu/		
City:	Durham	**State:**	NC
Zip Code:	27708	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	04/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1583		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The aim of this project is to understand the molecular bases for evolutionary changes in transcriptional regulation. We are focusing on one of the best-characterized cis-regulatory regions of any eukaryotic gene, the Endo16 locus of camarodont sea urchins. The project has three basic aims: (1) to identify the genetic and molecular changes that have produced functional compensation in promoter function among species, both on a small scale and a more extensive scale; (2) to identify the genetic and molecular changes that have produced a functional difference in transcription profiles among species; and (3) to identify evolutionary changes in promoter function at a biochemical level by testing known and artificial variants of transcription factor binding sites.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Understanding how gene regulatory systems operate and evolve is an essential part of several important Earth benefits. First, it will be fundamental to our understadning of the history of life on Earth, since it provides insights into how genetic systems evolve and what constraints they operate under. Second, it will provide insights into our understanding of the general properties of life: all life requires control systems to extract information from the genetic repository and deploy it in response to the environment.
Task Progress:	The basic aim of this project is to understand the molecular bases for evolutionary changes in transcriptional regulation. We are focusing on one of the best-characterized cis-regulatory regions of any eukaryotic gene, the Endo16 locus of camarodont sea urchins. The project has three basic aims: (1) to identify the genetic and molecular changes that have produced functional compensation in promoter function among species, both on a small scale and a more extensive scale; (2) to identify the genetic and molecular changes that have produced a functional difference in transcription profiles among species; and (3) to identify evolutionary changes in promoter function at a biochemical level by testing known and artificial variants of transcription factor binding sites. Major findings and results from year 1 of funding: Related to aim (1), we have determined that functional compensation in Endo16 transcription among the species Strongylocentrotuys purpuratus and Lytechinus variegatus is based on both cis (within promoter) and trans (transcription factor expression and/or function) components. The differences in promoter sequences between these species are dramatic: although the proximal 15% of the promoter is highly conserved, sequence similarity throughout the rest of the promoter is essentially at random levels. We showed that despite these sequence differences, the promoters of each species functions nearly perfectly in the other species. These results provide one of the most dramatic and clearly documented cases of functional compensation in promoter sequences yet identified, and suggest that mechanisms transcriptional regulation can evolve faster than the transcription profiles they generate. This work has been published (Romano and Wray 2003). We have not yet started on experiments related to aim (2). Related to aim (3), we are in the process of refining the techniques that will be used and have begun to gather data. During this first year of funding, we have also completed work on two spin-off projects: analyses of natural selection on transcriptional regulatory sequences in bacteria (Hahn et al. 2003) and hunans (Rockman et al, 2003). Two additional projects are nearly complete, and will be submitted for publication within the next two months: an anlysis of the population genetics of Endo16 cis-regulatory sequences and an experimental study of Endo16 protein function. Finally, we have published two reviews on the evolution of transcriptional regulation, both of which highlight our NASA-funded research (Wray 2003; Wray et al. 2003). Major findings and results from year 2 of funding: During the second year of funding, we have very nearly completed gathering data for Aim 3 and have laid the groundwork for completing aim by developing methods and carrrying out proof-of-concept experiments for Aim 2. We have developed proficiency with a biochemical assay of protein:DNA interaction using electromobility shift assays. To date we have tested all 12 of the natural alleles within the endo16 promoter region that occur within transcription factor binding sites. We have also tested all of the 11 fixed sequences differences in transcription factor binding sites between S. purpuratus and its closest living relative, S. droebachiensis. The most surprising finding from these analyses is that most of the derived alleles and fixed differences result in an increase in binding affinity. Some are relative small affinity differences (<2-fold) but a few are larger (up to 20-fold). The representation of affinity gains over losses is highly significant statistically, and suggests that positive selection plays an important role in the origin and maintainence of functional genetic changes within cis-regulatory regions. This work is currently being written up for publication. Regarding Aim 3, we have developed a high-throughput method for introducing into embryos constructs for functional testing in transient expression assays. We have optimized a system for delivering DNA-coated gold particles into eggs using a Gene Gun (BioRad). We have also begun to build full-length and mosaic expression constructs for testing endo16 regulatory regions from different species to test in expression assays. We anticipate no substantive difficulties at this point in completing the basic goals of Aim 2 during the final year of funding. During this year of funding, I also co-edited a methods book for experimental approaches to studying development in non-vertebrate deuterostomes (Ettensohn, Wessel and Wray, 2004). Research planned for year 3 of funding: Aim 1: completed. Aim 3: refine affinity measurements to higher degree of precision. Write up results. Expand affinity measurements to additional species as time permits. Aim 2: this is the last remaining aim. We will carry out experiments aimed at understading the mechanistic basis for differences in promoter function, as described in the previous section. These will follow the experiments outlined in our original proposal, as informed by results we have gained from carrying out Aims 1 and 2.
Bibliography Type:	Description: (Last Updated: 08/31/2006)
Articles in Peer-reviewed Journals	Balhoff, J. P.; Wray, G. A. "Evolutionary analysis of the well-characterized endo16 promoter reveals substantial variation within functional sites" Proceedings of the National Academy of Sciences (U.S.A.), in press , Jan-2005
Articles in Peer-reviewed Journals	Hahn, M. W.; Rockman, M.V.; Soranzo, N.; Goldstein, D. B.; Wray, G.A. ". Population genetic and phylogenetic evidence for positive selection on regulatory mutations at the Factor VII locus" Genetics 167: 867-877 , Jan-2004
Articles in Peer-reviewed Journals	Rockman, M.V.; Hahn, M. W.; Soranzo, N; Loisel, D.; Goldstein, D. B.; Wray, G. A. "Positive selection on MMP3 regulation has shaped heart disease risk" Current Biology 14: 1531-1539 , Jan-2004